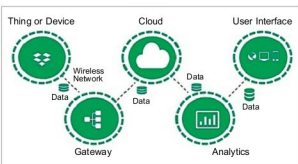


ments. Batteries used in these systems are expected to last up to 20 years without requiring much manual intervention. The uniqueness of IoT design elements is their ability to sleep when not in use.

Gaurav Sareen, country head-India, Sigfox, explains, "The transceiver wakes up only when it is supposed to transmit data. As soon as it is done, it goes back to sleep. So, the hardware is essentially asleep most of the time. That is how it conserves energy."

IoT systems consume power in two modes: dynamic (when the hardware is active) and static (when the hardware is asleep or in standby mode). It is essential to keep these factors in check. Consumption of both the MCU and the transceiver are considered. 32-bit MCUs can transmit data faster. Therefore their dynamic consumption can be much less, enabling the chip to save more energy. FPU compatibility further enhances efficiency. Large-



Key IoT components (Image source: rfpag.com)

scale applications in businesses can greatly benefit from FPU-supported MCUs.

Software for security. Security solutions are essential to ensure that no part of the system (hardware, software or network gateway level) is compromised.

Sareen mentions, "Secure Elements (SE) are in use to ensure safety at hardware level. An SE itself is an additional hardware component that encrypts data and checks authenticity of the device receiving it. The network layer is secured by

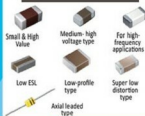
hiding the data in motion with advanced encryption schemes (AES). Data cannot be breached while being transmitted through network gateways. Finally, data itself is tested in the application layer to ensure it is infection-free." Authentication levels, including passwords and biometrics, provide secured access to all components and data. In addition, there are debug software that ensure proper functioning of the IoT system.

Why it's the best time

The job of designers is becoming even simpler with manufacturers introducing integrated modules like MCU-cum-transceiver units that facilitate quicker IoT system design. Efficient software tools allow speedy programming of embedded systems. With improved technology and capacities, isn't this the best time to be an IoT designer? **BY**

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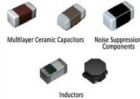
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